

COURSE TITLE : COMPOSITE LAB
COURSE CODE : 5099
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52/5
CREDITS : 2

TIME SCHEDULE

Module	Topics	Period
1	Testing of fibres	13
2	Determination of gel time, set and cure time	13
3	Production of composite products	13
4	Testing of composite products	13
TOTAL		52

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To determine denier, tex, tenacity, elongation at break and linear density of fibres.
2	2	To determine gel time, set time and cure time of the resin.
3	3	To produce composite products by hand lay up process and cast articles.
4	4	To carryout testing of composite products.

LIST OF EXPERIMENTS IN COMPOSITE LAB

MODULE I TESTING OF FIBRES.

- 1.1.0. To determine denier, tex, tenacity, elongation at break and linear density of fibres.
 - 1.1.1. Determine Denier and Tex of Cotton , Wool, Polyester, Glass and Coir fibres
 - 1.1.2. Find out the Tenacity of Cotton , Wool, Polyester, Glass and Coir fibres
 - 1.1.3. Determine the Linear density of Cotton and Polyester fibres

MODULE II DETERMINATION OF GEL TIME, SET AND CURE TIME

- 2.1.0. To determine gel time, set time and cure time of the resin.
 - 2.1.1. Determine the Gel time of Polyester resin
 - 2.1.2. Determine the Set time of Polyester resin
 - 2.1.3. Determine the cure time of Polyester resin

MODULE III PRODUCTION OF COMPOSITE PRODUCTS

- 3.1.0. To produce composite products by hand lay up process and cast articles.
 - 3.1.1. Perform the Hand lay up process for different products(single layer,double layer,inner and outer finished)
 - 3.1.2. Carry out Casting process with Polyester resin

MODULE IV TESTING OF COMPOSITE PRODUCTS

- 4.1.0. Testing of composite products
 - 4.1.1. Find out the Ignition loss of Composite material
 - 4.1.2. Determine the water absorption of composites
 - 4.1.3. Determine the Swelling of composites
 - 4.1.4. Determine the Density of composites
 - 4.1.5. Determination of void content in composite